

long
time scale
100 ms – s

Perception · World Model · Motion Planning

terrain recognition · human intent · obstacle prediction · foothold replanning

interaction state + $\dot{\theta}_{fall}$
→ replan / widen / slow / abort

walk command ·
footstep plan

short
time scale
1 – 20 ms

Interaction Dynamics Layer

External-wrench estimator

$$\hat{F}_{ext} = m \ddot{c}_{xy} - \sum_i F_{i,xy}$$

robot state: IMU · CoM vel
· foot wrenches

Interaction-confidence arbitration

1) confidence gate

$$g: |\hat{F}_{ext}| > f_{floor} + k_{\sigma} \sigma_f ?$$

2) persistence

p : transient ↔ sustained

$$u = g[(1 - p) u_{capture} + p u_{hold}]$$

velocity-command
bias u

actuation

robot state feedback
IMU, kinematics, foot wrenches

Robot

Whole-body realization

frozen RL locomotion policy